

The Tree ADT

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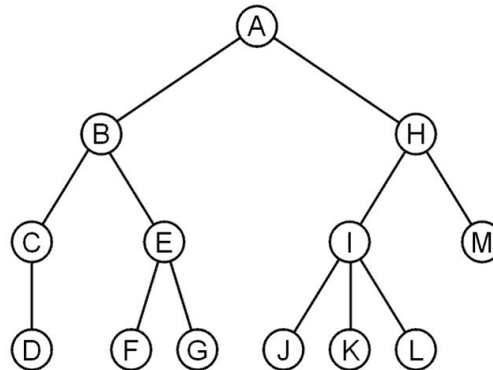


Trees

A rooted tree data structure stores information in nodes

Similar to linked lists:

- There is a first node, or root
- Each node has variable number of references to successors
- Each node, other than the root, has exactly one node pointing to it



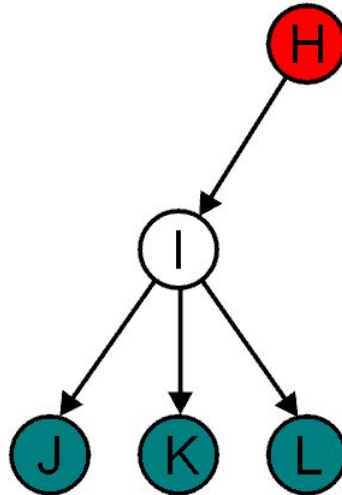
Tree Terminology

All nodes will have zero or more child nodes or children

- I has three children: J, K and L

For all nodes other than the root node, there is one parent node

- H is the parent I



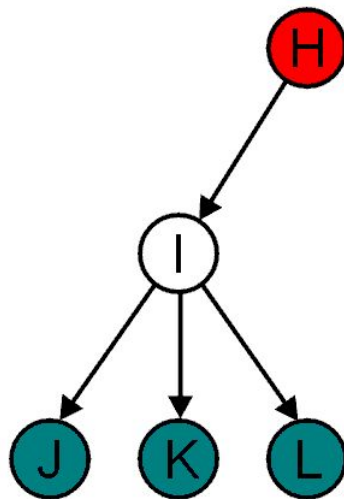
Tree Terminology

The degree of a node is defined as the number of its children:

- $\deg(I) = 3$

Nodes with the same parent are siblings

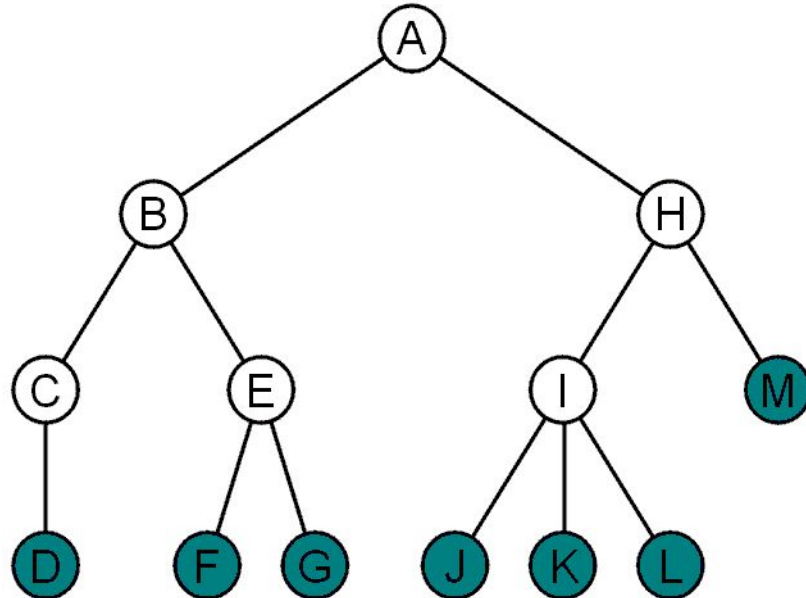
- J, K, and L are siblings



Tree Terminology

Nodes with degree zero are also called leaf nodes

All other nodes are said to be internal nodes, that is, they are internal to the tree



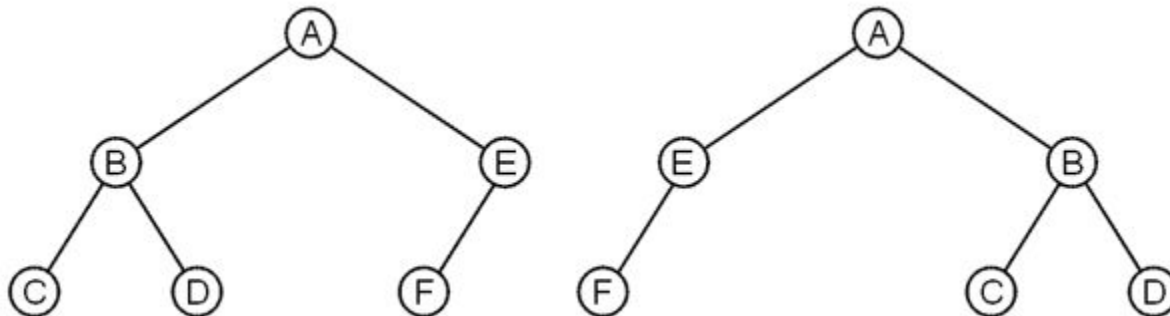
Tree Terminology

The trees below are considered equal if the order of the children is ignored

- unordered trees

They are different if order is relevant (ordered trees)

- We will usually examine ordered trees (linear orders)
- In a hierarchical ordering, order is not relevant



Tree Terminology

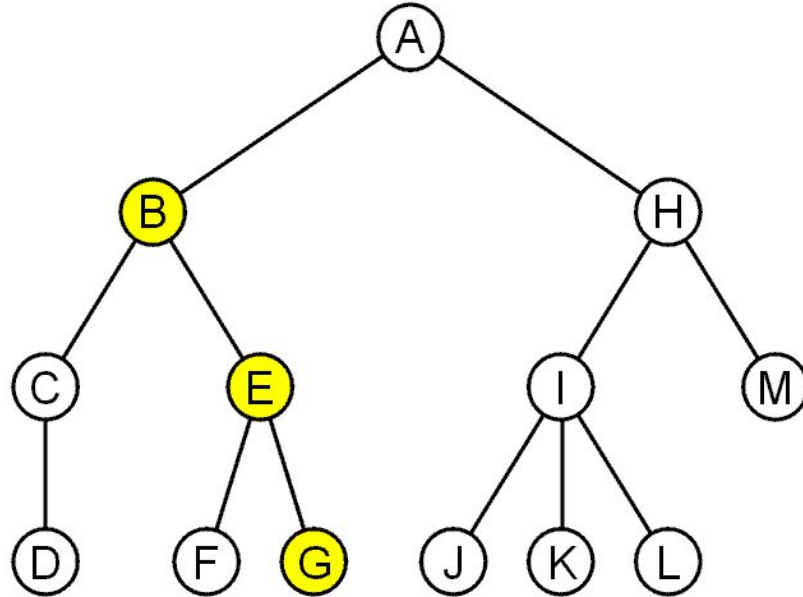
A path is a sequence of nodes

$$(a_0, a_1, \dots, a_n)$$

where a_{k+1} is a child of a_k is

The length of this path is n

E.g., the path (B, E, G)
has length 2

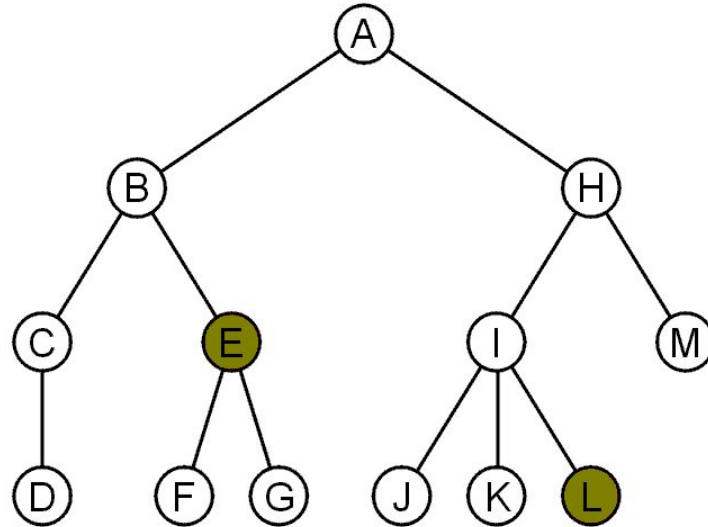


Tree Terminology

For each node in a tree, there exists a unique path from the root node to that node

The length of this path is the depth of the node, e.g.,

- E has depth 2
- L has depth 3



Tree Terminology

The height of a tree is defined as the maximum depth of any node within the tree

The height of a tree with one node is 0

- Just the root node

For convenience, we define the height of the empty tree to be -1

Example: XHTML and CSS

The XML of XHTML has a tree structure

Cascading Style Sheets (CSS) use the tree structure to modify the display of HTML

Example: XHTML and CSS

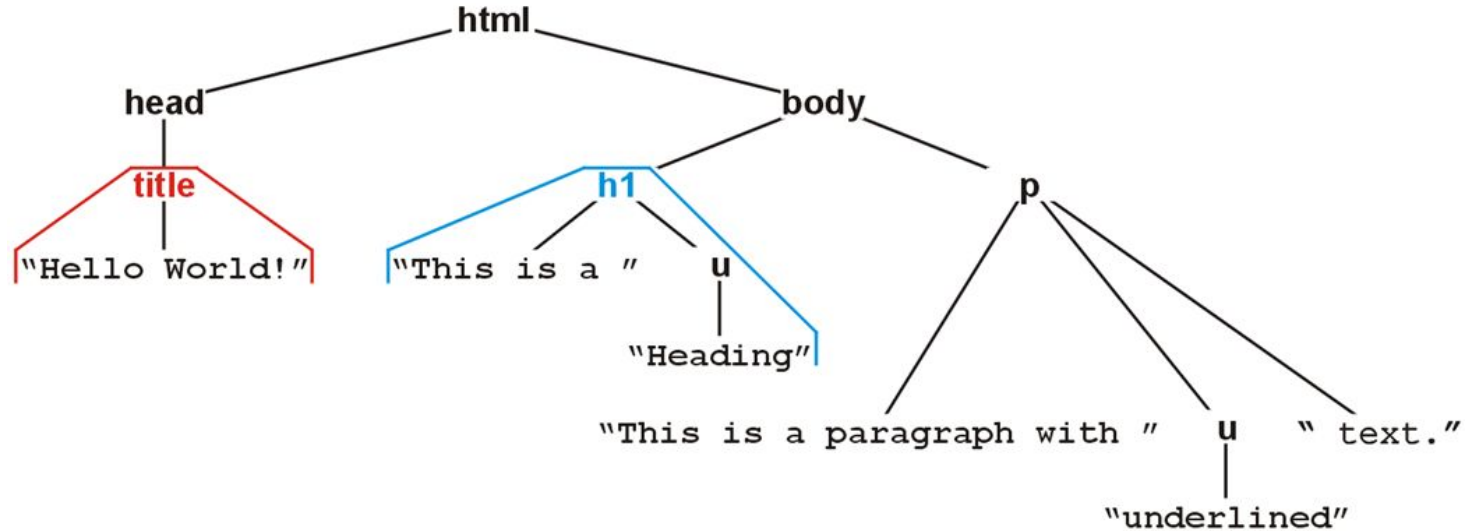
Consider the following XHTML document

```
<html>
  <head>
    <title>Hello World!</title>
  </head>
  <body>
    <h1>This is a <u>Heading</u></h1>

    <p>This is a paragraph with some
      <u>underlined</u> text.</p>
  </body>
</html>
```

Example: XHTML and CSS

The nested tags define a tree rooted at the HTML tag



Example: XHTML and CSS

Web browsers render this tree as a web page

